

Chahit Jain

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EDUCATION

University of Illinois Urbana-Champaign

August 2023-May 2027 (expected)

B.S. Computer Science + Statistics

GPA: 3.97 / 4.00

Coursework: Robot Manipulation, Mobile Robotics, Robotics Team Project, Deep Learning for Computer Vision, Applied Machine Learning, Basics of Statistical Learning, Data Structures, Algorithms, Computer Systems, Programming Languages & Compilers, Database Systems, Numerical Methods I.

Honors: Dean's List.

PUBLICATION AND PREPRINT

- Dwip Dalal, Shivansh Patel, **Chahit Jain**, Jeonghwan Kim, Utkarsh Mishra, Alex Baratian, Hyeonjeong Ha, Heng Ji, Svetlana Lazebnik, and Unnat Jain. "Generalizable VLA Finetuning via Representation Anchoring and Language-Action Alignment." arXiv preprint, 2026; ICLR 2027 submission in preparation.

RESEARCH & ENGINEERING EXPERIENCE

Visual and Embodied Intelligence (UnVEIL) Group

January 2026-Present

Undergraduate Researcher

- Led xArm7 data collection and policy deployment for Anchor-Align, including multi-camera RealSense calibration, action execution, and rollout analysis across VLA-Adapter and StarVLA.
- Implemented overlapping inference and action-chunk blending for smoother closed-loop execution; built a LIBERO-to-Blender pipeline for rendering recorded simulation trajectories.
- Evaluated four real-world distribution shifts with 20 rollouts per condition and analyzed failures using a six-mode taxonomy. Across three shared conditions, the team study reports mean success gains from 28.3% to 54.2% and from 36.7% to 60.0%.
- Currently developing inference-time guidance methods for VLA policies, with large-scale simulation experiments and real-world evaluation on an xArm7.

CS 498: Robotics Team Project

August 2026-Present

Course Assistant, University of Illinois Urbana-Champaign

- Develop F1TENTH and Booster K1 labs, set up and debug physical robot platforms, support SLAM instruction, and build reinforcement-learning assignments.

Language-Conditioned Agricultural Navigation (WaypointGen)

December 2025-May 2026

Independent Study Researcher, Distributed Autonomous Systems Lab

- Built the global navigation pipeline, extracting crop-row structure and spatial graphs from overhead imagery and grounding farmer-style instructions to explicit rows, trees, and waypoints.
- Implemented row and lattice recovery, missing-tree reasoning, and automated instruction generation and filtering; explored constrained, quantized Qwen-family inference on Jetson Orin.

NCSA SPIN / CyberAg

September 2024-August 2025

Undergraduate Researcher and Robotics Engineer

- Designed and fabricated a rocker-bogie rover for narrow crop rows, including the chassis and custom motor mounts in Fusion 360, then integrated motors, microcontrollers, Raspberry Pi compute, power systems, and sensing.
- Developed ROS 2 and Gazebo autonomy for obstacle avoidance, coverage, and formation; integrated ultrasonic and IMU sensing with Kalman filtering and evaluated row tracking and obstacle avoidance on prototype hardware.
- Ran repeated under-canopy LoRa experiments and prototyped anchor-based TDoA localization for GPS-degraded operation.

AI and Computer Vision Engineer

- Built a crowded multi-animal perception pipeline using YOLO, DeepSORT, and pose estimation; curated 2,146 annotated frames from 38 videos with approximately 22,000 animal instances.
- Developed photometric and GAN-based day-to-night augmentation and tested ONNX inference pipelines on NVIDIA Jetson and Raspberry Pi.

Parasol Lab

Summer 2026-Present

Undergraduate Researcher

- Integrated and modified the C++ Guided-PIBT planner within a Python experiment harness, resolved planner-simulator semantic mismatches, and added deterministic correctness checks.
- Evaluated corridor-aware decomposition with 84 physical channels and 21 node-disjoint paths; rejected a 203-region hypothesis after the gains failed to generalize across settings.

SELECTED ROBOTICS PROJECTS

Long-Horizon Manipulation with Imitation Learning

RoboCasa / robosuite course project; simulation

- Built a 6-DoF teleoperation and data-collection workflow in RoboCasa with task-stage checkpoints, simulator-state restoration, subtask retries, and corrective intervention.
- Expanded 15 source demonstrations into 1,887 MimicGen trajectories, retained 1,000 after filtering, and trained a BC-RNN policy that reached approximately 97% success across 100 static-task evaluation episodes.

F1TENTH Autonomous Racing

Course project; physical 1/10-scale racecar

- Implemented and tested a C++ autonomy stack spanning PID wall following, Pure Pursuit, RRT/RRT*, SLAM-based planning, trajectory optimization, and Follow-the-Gap.
- Diagnosed a repeatable physical-car failure by tracing unstable and missing LiDAR returns to a phantom free-space gap rather than a planner-logic defect.

PROFESSIONAL SERVICE

- Reviewer, Conference on Robot Learning (CoRL), 2026

TECHNICAL SKILLS

Programming and systems: C++, Python, C, Linux, Git, Docker, HTCondor, SLURM.

Robotics and hardware: ROS 2, UFactory xArm7, F1TENTH, Booster K1, Intel RealSense, Jetson Orin, Raspberry Pi, LiDAR, IMU and ultrasonic sensing, LoRa, Fusion 360.

Robot learning and evaluation: PyTorch, vision-language-action policies, behavior cloning, BC-RNN, MimicGen, GELLO teleoperation, camera calibration, real-robot evaluation.

Autonomy and perception: SLAM, MPPI, Pure Pursuit, RRT/RRT*, OpenCV, YOLO, DeepSORT, ViTPose, HRNet, SLEAP, SAM.

Simulation and deployment: MuJoCo, Gazebo, robosuite, RoboCasa, LIBERO, ONNX, quantization.